Lec-5
10/5/20 5:10 PM

① number System
② Data - Types
③ Type casting

$3746 = 3\times10^3 + 7\times10^2 + 4\times10^1 + 6\times10^0$

$3746/10 = 374$
$374/10 = 37$
$37/10 = 3$
$3/10 = 0$

$1\rfloor0\,1\,1$
$\rfloor8\,4\,6\,7$
$\rfloor5\,3\,4\,7$
$1\,3\,9\,1\,0$

$[5\,2\,0\,3\,4] \times 10^1$ 3416
34655

① Binary → 2
② Decimal → 10
③ octal → 8 [0 to 7]

$10\%10 = 0$ $8\%10 = 8$
$10/10 = 1$ $8/10 = 0$
$11\%10 = 1$ $13\%10 = 3$
$11/10 = 1$ $13/10 = 1$

$1\,0\,1\,1$
$5\,2\,0\,3\,4$
$3\,4\,3\,5\,5$
$5\,0\,4\,3\,3$

$[1\,0\,0\,1\,1\,0\,1]$
$0\,1\,1\,0\,0\,1$
$1\,0\,1\,1\,0\,0$

$2\cdot12 = 0$
$2/2 = 1$
$1/2 = 1$
$1/2 = 0$
$3\%2 = 1$
$3/2 = 1$

$a = a|b$
$b = a\&b$
$a = a\&b$

$a\&b = a$

$c = a$
$b = 7$

while
$a = b$
$b = c$

$a = a|b = 16$
$b = a - b$
$= 17 - 7 = 9$
$a = a - b$
$16 - 9 = 7$

2	37
2	18
2	9
2	4
2	2
2	1
	0

$\%2$ 1, 0, 0, 1, 0, 1

$37 \to$ Binary

mul = 1
while (n > 0) {
sum = n % 2
sum = sum * mul
mul = mul * 10
n = n/2
}

100101 ✓
111001

2	37
2	18
2	9
2	4
2	2
2	1
	0

$\%2$ 1, 0, 0, 1, 0, 1

100000
100
1
100101
37

$sum = 1 + 0 + 100 + 0 + 0 + 100000$

$111001 = 1\times2^5 + 1\times2^4 + 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2^0$

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 37;
        int mul = 1;// 10^0
        int sum = 0;
        while (n > 0) {
            int rem = n % 2;
            sum = sum + rem * mul;
            mul = mul * 10;
            n = n / 2;
        }
        System.out.println(sum);
}
```

STC = 10
dec = 2

10	10110
10	1011
10	101
10	10
10	1
	0

$0 \times 2^0 = 0$
$1 \times 2^1 = 2$
$1 \times 2^2 = 4$
$0 \times 2^3 = 0$
$1 \times 2^4 = 16$
22

mul = 1
while (n > 0) {
int rem = n % dec
sum = sum + rem × mul
mul = mul × 10
n = n / dec
}

4 → 7
10 →

data Types

Primitive
Non Primitive ✗

byte
short
int
long
float
double

char
boolean

data	Size	Range
byte	1 byte / 8 bit	
short	2 byte / 16 bit	
int	4 byte / 32 bit	
long	8 byte / 64 bit	
float	4 byte / 32 bit	
double	8 byte / 64 bit	
char	2 byte / 16 bit	
boolean	1 bit	

$8b \to 1 byte$
boolean → true

$10 \times 10 \times 10 \to 1000$
$10 \times 10 \times 8 \to 800$
$1024 \to 1$ kb
$1024 \times 8 \to 8192$ bit

byte : [0 0 0 0 0 0 0 0] [0, 255]

① Sign-bit
+ve
-ve
[-128, 127]
[0, 255]
-128
128
250
255
-127
1000 0000
0000 0000

1's complement

10101101
01010010
72
01110100
10001011 (-72)
10
00000000
1111

83
$1001\,0011$
$+\ 0110\,1100$
$0110\,1010$
2's complement
$1000\,0000$
128

Range
$-128, 127]$

+ve ← 0 1 1 1 1 1 1 1
-ve ← 1 0 0 0 0 0 0 0

$0\,111\,1111$
$0\,00\,0001$
$1000\,0000$

$2\cdot2 = 0$
$2/2 = 1$

PA

byte → 8 bit → $-128 \to +127 \to -2^7$ to $2^7 - 1$
Short → 16 bit → $\to -2^{15}$ to $2^{15} - 1$
int → 32 bit → -2^{31} to $2^{31} - 1$
long → 64 bit → -2^{63} to $2^{63} - 1$

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        byte b = 5;
        short s = 5;
        int i = 5;
        long l = 5;
}
```

$0000\,0101$
101
$2 \mid 5$
$2 \mid 2$
$2 \mid 1$
0

$1\,0\,1\,0\,1\,1\,0\,0$
$1\,0\,1\,0\,1\,1\,0\,0$
$0\,1\,0\,1\,0\,0\,1$
$0\,1\,0\,1\,0\,0$
-84

$1\,0\,0\,1\,0\,1\,1\,0\,0$
s = 5
$00\,10\,1100$
$2^7\ 2^6\ 2^5\ 2^4\ 2^3$
$32 + 8 + 4 = 44$

byte b = 5
$0000\,0101$
byte b = -5
$1111\,1011$
-5
-128
-2^{31} to $2^{31} - 1$
-2^{63} to $2^{63} - 1$

19 - c
21 c
5
a, e, i, o, u
'a' 'e' 'i' 'o' 'u'
A B C
'0' '1' '2'
255